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1 *****
2 * TS-2068 LOW LEVEL UNIFIED COMMUNICATION PROTOCOL VERS: 2.5 SEP-2023 *
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13 *****
14      **** PRELIMINAR ****
15      **** AND ****
16      **** CONFIDENTIAL ****
17 *****
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21 INTRODUCTION

23 This document details the communication protocol specifications between
24 the operating system of the TS-2068 computer and the peripherals that
25 support functionalities of storage and communication.

27 The new basic commands added to the traditional ones use the protocol
28 TPI ("Timex Protocol Interface"). In this document you will find examples
29 of these extended commands in Basic (called TPI commands from now on) and
30 the detail of the low-level protocol data exchange for each one of them.

32 The information herein is confidential, only the people authorized may use
33 it in order to develop the interfaces and firmware required, and must not
34 be distributed or published to third parties without the previous express
35 author's consent.

37 The author is not responsible for any damage or harm caused by the use of
38 the information contained at this document. However, if you find any error
39 or consider that a functionality should be added or improved, please
40 contact me at panegustavo@yahoo.com.ar with the subject TPI ERROR
41 or TPI CHANGE REQUEST (these observations will be analyzed and could
42 be included at further releases of this document).

44 This document supersedes all previous versions and may change without notice.

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140 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 2
141
142 ;-----
143 ; PICO INTERFACE BIOS
144 ;-----
145
146 -----
147 EXROM
148 ADDRESS LABEL FUNCTIONALITY
149 -----
150 01840H G_MODE Get Output Mode (Return Mode in BC (01=SD / 00=TAPE )
151 -----
152 01842H S_MODE Set Output Mode in A A=0 Select TAPE Port
153 A=1 Select SD Port
154 -----
155 01844H G_VERS Get Bios Version (Return Version in BC)
156 -----
157 01846H TX_A Transmit Character in A
158 -----
159 01848H RX_A Receive Character in A
160 -----
161 0184AH C_END Command END C_END Return Z No Error, NZ Error Code in A
162 Return Cy Error I/O
163 -----
164 0184CH WF_NPH Wait Command Execution ( Cy = 1 Timeout ERROR)
165 -----
166
167 ++++++
168 THE TS-PICO BASIC SYSTEM USE HOME MEMORY FROM THE THE LOCATIONS 5DCD TO 5DDC.
169 THIS AREA MUST BE SET TO 00H ONLY DURING BOOT TIME. (TIMEX 2068 INITIALIZE THIS AREA)
170 The addresses atarting from 5DDD to 633F Are reserved for future new software releases.
171 ++++++
172
173 -----
174 GET STATUS SAMPLE PROGRAM
175 -----
176
177 WR-DAT EQU 00EH ; PICO DATA PORT FOR READ OPERATION
178 RD-DAT EQU WR-DAT ; PICO DATA PORT FOR WRITE OPERATION
179 RD-STA EQU WR-DAT+1; PICO STATUS PORT FOR I/O OPERATION
180
181 DKHSPT EQU 0F4H
182 HREXPT EQU 0FFH
183 FS_GS EQU 000H ; File System Get Status Command
184
185 AS_CMD EQU 041H ; Assembler CMD
186 DA_BLK EQU 044H ; Data Block Type
187 FS_CMD EQU 050H
188 CMD_OK EQU 0A1H
189 BL_ERR EQU 0BEH ; Block Error
190 CMD_END EQU 001H ; Command END
191
192 TX_A EQU 01846H ; TS-PICO SEND BYTE LOW LEVEL FUNCTION
193 RX_A EQU 01848H ; TS-PICO RECEIVE BYTE LOW LEVEL FUNCTION
194 WF_NPH EQU 0184CH ; TS-PICO WAIT FOR EXEC. COMPLETE LOW LEVEL FUNCTION
195
196 BLKSIZE EQU 256 ; Filesystem Block Size
197 BUFMAX EQU BLKSIZE ; Maximun Block Size (USE FOR FUNCTIONS DATA)
198
199 ;-----
200 CALL CLSBUF ; Clear Buffers Area
201
202 LD A,FS_GS ; File System Get Status Command
203 CALL EXEC ; Execute Function
204 RET C ; Return if Error
205 ;-----
206 ; Process BUFDAT here
207 ;-----
208 RET
209
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210 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 3
211
212 ;-----TS-PICO LOW LEVEL COMMUNICATION INTERFACE-----
213 EXEC:  DI
214         LD      (FUNCT),A      ; Set Function to use
215         IN      A,(DKHSPT)
216         LD      (SA_HS),A      ; Save Original Value
217         LD      A,1            ; Disable Home Bank
218         OUT     (DKHSPT),A      ; in Chuncck 0
219         IN      A,(HREXPT)
220         LD      (SA_DK),A      ; Save Original Value
221         SET     7,A
222         OUT     (HREXPT),A      ; Select EXROM BANK
223
224         PUSH    BC              ; Save Register
225         PUSH    DE
226         PUSH    HL
227
228         LD      HL,PREHDR
229         LD      D,0            ; Reset CRC Header
230         LD      B,9            ; PRE_HDR LEN
231
232 LOOPT:  LD      A,(HL)          ; Get Byte to TX
233         CALL    TX_A            ; Send Command
234         XOR     D               ; Calculate CRC
235         LD      D,A            ; Save CRC
236         DJNZ    LOOPT
237
238         CALL    TX_A            ; Send CRC FROM TIMEX
239         CALL    RX_A            ; Receive Header Receive Status
240         CP      CMD_OK          ; Check if Header Was Received OK
241         JR      NZ,ERROR        ; Jump to Report ERROR in CMD
242         CALL    WF_NPH          ; Wait For Command Execution Ready
243
244         CALL    RX_A            ; Receive Block Type
245         CP      DA_BLK
246         JR      NZ,ERRBE        ; Jump to Report ERROR in CMD
247         LD      D,A            ; Save CRC in D
248
249         CALL    RX_A            ; Receive Block Type
250         LD      C,A             ; Get Block LEN LSB
251         XOR     D               ; Calculate CRC
252         LD      D,A            ; Save CRC in D
253
254         CALL    RX_A            ; Receive Block Type
255         LD      B,A             ; Get Block LEN MSB
256         XOR     D               ; Calculate CRC
257         LD      D,A            ; Save CRC in D
258
259         OR      C               ; if A=0 NO DATA AVAILABLE
260         JR      Z,ERRBE        ; Jump to Report ERROR in CMD
261
262 LCDRX:  LD      HL,BUFDAT        ; HL=Received Buffer
263         CALL    RX_A            ; Receive Command Data
264         LD      (HL),A          ; Store in Buffer
265         INC     HL              ; Increment Memory Pointer
266         XOR     D               ; Calculate CRC
267         LD      D,A            ; Save CRC in D
268         DEC     BC
269         LD      A,B
270         OR      C               ; Test if Block END
271         JR      NZ,LCDRX
272
273         CALL    RX_A            ; Get CRC
274         XOR     D               ; Calculate CRC
275         JR      NZ,ERRBE        ; Jump to Report ERROR in CMD BLOCK
276
277         LD      A,CMD_END
278         CALL    TX_A            ; Send Command
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280 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 4

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281
282     LD      A,(BUFDAT)      ; Get COMMAND EXECUTION STATUS
283     AND     A               ; Set Cy = 0
284     CP      CMD_OK          ; Test if Execution was OK
285     JR      Z,ERROR         ; Return Cy = 0 NO_ERROR
286 ;-----
287 ERRBE: LD      A,BL_ERR      ; Return Block Error
288     SCF                     ; Set Cy=1 Indicating CMD Error
289 ;-----
290 ERROR: CALL    EXIT
291
292     POP     HL
293     POP     DE
294     POP     BC
295     EI
296     RET                     ; Return Cy = 1 IF ERROR
297 ;-----
298 EXIT:  PUSH    AF           ; Save AF
299     LD      A,(SA_HS)       ; Get Original Value
300     OUT     (DKHSPT),A      ; Set Original DOCK Port Config
301     LD      A,(SA_DK)       ; Get Original Value
302     OUT     (HREXPT),A      ; Set Original Port Config
303     POP     AF             ; Restore AF
304     RET
305 ;-----
306 CLSBUF: PUSH    HL
307     PUSH    BC
308     PUSH    DE
309     LD      HL,PREHDR
310     XOR     A
311     LD      (HL),A
312     LD      BC,BUF_END-FUNCT
313     LD      DE,PREHDR+1
314     LDIR
315     LD      A,AS_CMD
316     LD      (CMD),A
317     LD      A,FS_CMD
318     LD      (SUBSID),A
319     POP     DE
320     POP     BC
321     POP     HL
322     RET
323 ;-----
324 SA_HS:  DEFB    0
325 SA_DK:  DEFB    0
326 ;-----
327 PREHDR:
328
329 CMD:    DEFB     AS_CMD      ; BLOCK TYPE = ASSEMBLER
330 SUBSID: DEFB     FS_CMD      ; SUBSYSTEM ID = FILE SYSTEM
331 FUNCT:  DEFB     00h        ; FUNCTION ID
332 FMODE:  DEFB     00h        ; FUNC_MODE
333 AUXP:   DEFB     00h        ; AUXILIARY PARAM
334 PMR_0:  DEFB     00h        ; PARAMETER 0
335 PMR_1:  DEFB     00h        ; PARAMETER
336 PMR_2:  DEFB     00h        ; PARAMETER
337 PMR_3:  DEFB     00h        ; PARAMETER
338 CRC_RX: DEFB     00h        ; CRC FRAME RECEIVED
339 ERR_RX: DEFB     00h        ; ERROR CODE RECEIVED
340 ;-----
341 BUFDAT: DEFS     BUFMAX+256,0 ; Minimun SPACE is 384 Bytes
342                                     ; Determined By Function Response Len
343                                     ; Received Buffer 512 (SIZE=BUFMAX + 256)
344 BUF_END:
345 ;-----
346 BUFADD: DEFS     4096,0       ; Extra Space Received Buffer
347                                     ; Defined Hera (Use for DIR-LISTING Func.)
348 BUF_ADD:
349 ;-----
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350 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 5

351
352 -----
353 TS-PICO TPI PROTOCOL USE FOR THE BASIC LANGUAGE TPI EXTENSION
354 -----
355

356 SAVE "TPI:RESET"

357 -----

358
359 =====
360 BYTE STAT DATA DATA BYTE DESCRIPTION
361 NUMB IN OUT IN
362 ----
363 PORT RD-STA WR-DAT RD-DAT Z80 I/O PORT
364 -----
365 00 0x42 BLOCK TYPE = ("B" = BASIC)
366 01 0x00 FUNCTION = (00 = SAVE)
367 02 0xFF BANK = (FF = HOME)
368 03 0x00 SESSION ID (Low Byte)
369 04 0x00 SESSION ID (High Byte)
370 05 0x00 MEMORY ADDRESS (Low Byte)
371 06 0x00 MEMORY ADDRESS (High Byte)
372 07 0x09 DATA BLOCK LEN (Low Byte)
373 08 0x00 DATA BLOCK LEN (High Byte)
374 -----
375 09 0xB4 CRC FROM TIMEX
376 -----
377 10 0x01 Read Status Code (01 = No Error)
378 -----WAIT EXECUTION 2.8ms-----
379 0x40 Read CONTINUE FLAG (BIT 6=1)
380 -----
381 11 0x44 BLOCK TYPE = ("D" DATA BLOCK CMD)
382 12 0x09 COMND LEN L
383 13 0x00 COMND LEN H
384 -----
385 14 0x54 "T"
386 15 0x50 "P"
387 16 0x49 "I"
388 17 0x3A ":"
389 18 0x52 "R"
390 19 0x45 "E"
391 20 0x53 "S"
392 21 0x45 "E"
393 22 0x54 "T"
394 -----
395 23 0x6F CRC FROM TIMEX
396 -----WAIT EXECUTION 2.8ms-----
397 0x40 Read CONTINUE FLAG (BIT 6=1)
398 -----
399 24 0x01 Status Code (No Error)
400 -----
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420 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 6

421
422 -----
423 TS-PICO TPI PROTOCOL USED FOR FILE SYSTEM EXTENSIONS
424 -----
425

426 -----
427 PRE HEADER ID SETTINGS
428 -----

429
430 0x42 = "A" (ASM COMND) *** DIRECT I/O CMD TO THE TS-PICO ***
431 0x50 = FILE_SYSTEM_CMD
432

433 -----
434 ERROR CODES ON FUNCTION RETURN
435 -----

436
437 CODE EXECUTION VALUES & ERROR

438 ;-----
439 0x01 END_CMD HOST FINISH THE CMD EXCHANGE MSG - OK -
440 ;-----
441 0xA0 INVALID I/O DEVICE
442 0xA1 CMD OK - NO ERROR
443 0xA2 MEDIA REMOVED
444 0xA3 MEDIA NOT RECOGNIZED
445 0xA4 MEDIA WAS CHANGED
446 0xA5 MEDIA FAILURE
447 0xA6 MEDIA IS WRITE PROTECT
448 0xA7 MEDIA SET R/O MODE AUTOMATICALLY
449 0xA8 MEDIA IS FULL
450 0xA9 MEDIA IS EMPTY
451 0xAB INVALID FILE NAME
452 0xAC FILE IS WRITE PROTECTED
453 0xAD FILE IS NOT OPENED
454 0xAE FILE IS NOT CLOSED
455 0xAF FILE R/W OPERATION OUT OF RANGE
456 0xB0 FILE R/W POINTER AT END OF FILE
457 0xB1 INVALID FUNCTION ID
458 0xB2 INVALID SESSION ID
459 0xB3 INVALID USER ID
460 0xB4 INVALID RECORD NUMBER
461 0xB5 INVALID BLOCK SIZE
462 0xB6 PARAMETER IS TOO BIG
463 0xB7 PARAMETER INVALID
464 0xB8 INTERNAL ERROR - STOP
465 0xB9 INVALID DIRECTORY NAME
466 0xBA NO DIRECTORY FOUND
467 0xBB NO DIRECTORY SPACE
468 0xBC DIRECTORY IS WRITE PROTECT
469 0xBD DIRECTORY IS NOT EMPTY
470 0xBE BLOCK ERROR
471 0xBF DISK NOT MOUNTED
472 0xC0 FUNCTION FAIL - UNKNOWN ERROR
473 ;-----
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490 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 7
491
492 -----
493 GET STATUS COMMAND = 00          - GET STATUS INFORMATION FROM STORAGE DEVICE
494 -----
495
496 =====
497 BYTE   STAT   DATA   DATA   BYTE DESCRIPTION
498 NUMB   IN     OUT     IN
499 ----   -
500 PORT   RD-STA WR-DAT  RD-DAT  Z80 I/O PORT
501 =====
502 00              0x41          = "A" (ASM COMND) *** DIRECT I/O CMD TO THE TS-PICO ***
503 01              0x50          = SUBSYSTEM ID (50 = FILE_SYSTEM_CMD)
504 02              0x00          = FUNCTION ID (00 = GET STATUS)
505 03              0x00          = FUNC MODE      NOT USED
506 04              0x00          = RESERVED      NOT USED
507 05              0x00          = PMR_0         NOT USED
508 06              0x00          = PMR_1         NOT USED
509 07              0x00          = PMR_2         NOT USED
510 08              0x00          = PMR_3         NOT USED
511 -----
512 09              0x11          CRC FROM TIMEX (from bytes 00 to 08)
513 -----
514 10              0xA1          Read Status Code (A1 = No Error)
515 -----
516 -----WAIT EXECUTION 2.8ms-----
517 -----
518 11              0x40          Read CONTINUE FLAG (BIT 6=1)
519 -----
520 12              0x44          "D" (DATA BLOCK CMD)
521 13              0x10          BLOCK LEN L
522 14              0x01          BLOCK LEN H
523 -----
524 15              0xA1          COMMAND EXECUTION STATUS (0xA1 = OK - NO ERROR)
525 -----
526 16              0x00          MOUNT_READY_FLAGS
527 17              0x00          SESION ID
528 18              0x00          USER ID
529 19              0x00          BLOCK SIZE LSB
530 20              0x01          BLOCK SIZE MSB
531 21              0x00          CURRENT FILE ACCESS MODE
532 22              0x00          CURRENT FILE OPEN MODE
533 23              0x00          PENDING WRITE BLOCKS
534 24              0x00          CURRENT POINTER BLOCK NUMBER LSB
535 25              0x00          CURRENT POINTER BLOCK NUMBER MSB
536 26              0x00          CURRENT POINTER EXTENT NUMBER
537 27              0x00          CURRENT FILE SIZE (LOW BYTE) (SIZE IN MB)
538 28              0x00          CURRENT FILE SIZE (MID BYTE)
539 29              0x00          CURRENT FILE SIZE (HIGH BYTE)
540 30              0xA1          FILESYSTEM_LAST_ERROR
541 31              0x00          COMMUNICATION ERROR COUNTER LSB
542 32              0x00          COMMUNICATION ERROR COUNTER MSB
543 33              0x00          FILE I/O ACCESS ERROR COUNTER LSB
544 34              0x00          FILE I/O ACCESS ERROR COUNTER MSB
545 35              0x00          HARDWARE ERROR ( 00 = NORMAL)
546 36              0x10          MAXIMUN FILES TO OPEN ( 10H = 16 FILES OPENEND)
547 37              0x01          TOTAL FILES OPEN ( 00 = NONE)
548 -----
549 38              0x04          CURRENT_PATH_LEN ( 00 = NO MOUNTED FILE)
550 39              0x5C          "\"
551 40              0x44          "D"
552 41              0x53          "S"
553 42              0x4B          "K"
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560 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 8
561
562 =====
563 BYTE      STAT      DATA      DATA      BYTE DESCRIPTION
564 NUMB      IN        OUT        IN
565 -----
566 PORT      RD-STA    WR-DAT      RD-DAT      Z80 I/O PORT
567 =====
568 43                                     0xE1      CURRENT_DISK_INFO_LEN LSB (00 = IF NO INFO AVAILABLE)
569 44                                     0x00      CURRENT_DISK_INFO_LEN MSB
570 -----
571 45                                     0x43      "C"
572 46                                     0x61      "a"
573 47                                     0x72      "r"
574 48                                     0x64      "d"
575 49                                     0x20      " "
576 50                                     0x69      "i"
577 51                                     0x73      "s"
578 52                                     0x20      " "
579 53                                     0x70      "p"
580 54                                     0x72      "r"
581 55                                     0x65      "e"
582 56                                     0x73      "s"
583 57                                     0x65      "e"
584 58                                     0x6E      "n"
585 59                                     0x74      "t"
586 60                                     0x2E      "."
587 61                                     0x0D      " "
588 62                                     0x0A      " "
589 63                                     0x43      "C"
590 64                                     0x61      "a"
591 65                                     0x72      "r"
592 66                                     0x64      "d"
593 67                                     0x20      " "
594 68                                     0x74      "t"
595 69                                     0x79      "y"
596 70                                     0x70      "p"
597 71                                     0x65      "e"
598 72                                     0x3A      ":"
599 73                                     0x20      " "
600 74                                     0x20      " "
601 75                                     0x20      " "
602 76                                     0x20      " "
603 77                                     0x20      " "
604 78                                     0x20      " "
605 79                                     0x20      " "
606 80                                     0x20      " "
607 81                                     0x20      " "
608 82                                     0x53      "S"
609 83                                     0x44      "D"
610 84                                     0x48      "H"
611 85                                     0x43      "C"
612 86                                     0x0D      " "
613 87                                     0x0A      " "
614 88                                     0x43      "C"
615 89                                     0x6C      "l"
616 90                                     0x75      "u"
617 91                                     0x73      "s"
618 92                                     0x74      "t"
619 93                                     0x65      "e"
620 94                                     0x72      "r"
621 95                                     0x73      "s"
622 96                                     0x3A      ":"
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700 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 10

701	=====				
702	BYTE	STAT	DATA	DATA	BYTE DESCRIPTION
703	NUMB	IN	OUT	IN	
704	----	----	----	----	
705	PORT	RD-STA	WR-DAT	RD-DAT	Z80 I/O PORT
706	=====				
707	151			0x20	" "
708	152			0x20	" "
709	153			0x20	" "
710	154			0x20	" "
711	155			0x20	" "
712	156			0x20	" "
713	157			0x33	"3"
714	158			0x31	"1"
715	159			0x30	"0"
716	160			0x39	"9"
717	161			0x39	"9"
718	162			0x39	"9"
719	163			0x30	"0"
720	164			0x34	"4"
721	165			0x0D	" "
722	166			0x0A	" "
723	167			0x42	"B"
724	168			0x6C	"l"
725	169			0x6F	"o"
726	170			0x63	"c"
727	171			0x6B	"k"
728	172			0x20	" "
729	173			0x53	"S"
730	174			0x69	"i"
731	175			0x7A	"z"
732	176			0x65	"e"
733	177			0x3A	": "
734	178			0x20	" "
735	179			0x20	" "
736	180			0x20	" "
737	181			0x20	" "
738	182			0x20	" "
739	183			0x20	" "
740	184			0x20	" "
741	185			0x20	" "
742	186			0x35	"5"
743	187			0x31	"1"
744	188			0x32	"2"
745	189			0x0D	" "
746	190			0x0A	" "
747	191			0x56	"V"
748	192			0x6F	"o"
749	193			0x6C	"l"
750	194			0x75	"u"
751	195			0x6D	"m"
752	196			0x65	"e"
753	197			0x20	" "
754	198			0x74	"t"
755	199			0x79	"y"
756	200			0x70	"p"
757	201			0x65	"e"
758	202			0x20	" "
759	203			0x69	"i"
760	204			0x73	"s"
761	205			0x3A	": "
762	206			0x20	" "
763	207			0x20	" "
764	208			0x20	" "
765	209			0x20	" "

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770 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 11

771	=====				
772	BYTE	STAT	DATA	DATA	BYTE DESCRIPTION
773	NUMB	IN	OUT	IN	
774	----	----	----	----	
775	PORT	RD-STA	WR-DAT	RD-DAT	Z80 I/O PORT
776	=====				
777	210			0x46	"F"
778	211			0x41	"A"
779	212			0x54	"T"
780	213			0x33	"3"
781	214			0x32	"2"
782	215			0x0D	" "
783	216			0x0A	" "
784	217			0x56	"V"
785	218			0x6F	"o"
786	219			0x6C	"l"
787	220			0x75	"u"
788	221			0x6D	"m"
789	222			0x65	"e"
790	223			0x20	" "
791	224			0x73	"s"
792	225			0x69	"i"
793	226			0x7A	"z"
794	227			0x65	"e"
795	228			0x20	" "
796	229			0x28	"("
797	230			0x4D	"M"
798	231			0x62	"b"
799	232			0x29	")"
800	233			0x3A	":"
801	234			0x20	" "
802	235			0x20	" "
803	236			0x31	"1"
804	237			0x35	"5"
805	238			0x31	"1"
806	239			0x38	"8"
807	240			0x35	"5"
808	241			0x0D	" "
809	242			0x0A	" "
810	243			0x56	"V"
811	244			0x6F	"o"
812	245			0x6C	"l"
813	246			0x75	"u"
814	247			0x6D	"m"
815	248			0x65	"e"
816	249			0x20	" "
817	250			0x46	"F"
818	251			0x72	"r"
819	252			0x65	"e"
820	253			0x65	"e"
821	254			0x20	" "
822	255			0x53	"S"
823	256			0x70	"p"
824	257			0x61	"a"
825	258			0x63	"c"
826	259			0x65	"e"
827	260			0x3A	":"
828	261			0x20	" "
829	262			0x31	"1"
830	263			0x34	"4"
831	264			0x32	"2"
832	265			0x38	"8"
833	266			0x30	"0"
834	267			0x0D	<ENTER CR>
835	268			0x0A	<LINE FEED>
836	269			0x00	END MARKER
837	-----				
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840 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 12
841
842 =====
843 BYTE      STAT      DATA      DATA      BYTE DESCRIPTION
844 NUMB      IN        OUT        IN
845 ----      -
846 PORT      RD-STA    WR-DAT      RD-DAT      Z80 I/O PORT
847 =====
848 -----FILE OPEN SECTION PATHNAME (FULL PATHNAME)-----
849 270                                0x0F      CURRENT_PATHNAME_LEN (00 = IF NO MORE FILE OPENED)
850 271                                0x5C      "\"
851 272                                0x44      "D"
852 273                                0x53      "S"
853 274                                0x4B      "K"
854 275                                0x5C      "\"
855 276                                0x43      "C"
856 277                                0x50      "P"
857 278                                0x4D      "M"
858 279                                0x2D      "-"
859 280                                0x32      "2"
860 281                                0x32      "2"
861 282                                0x2E      "."
862 283                                0x49      "I"
863 284                                0x4D      "M"
864 285                                0x47      "G"
865 -----
866 286                                0x00      LEN = 0 NO MORE FILES TO PRINT
867 -----
868 287                                0xA9      CRC FROM TS-PICO (from bytes 12 to 286)
869 -----
870 288                                0x01      Send END Command / ERROR CODE FROM TIMEX
871 -----
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980 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 14
981 -----
982 OPEN COMMAND = 02          - OPEN A FILE IN THE STORAGE DEVICE
983 -----
984 =====
985 BYTE   STAT   DATA   DATA   BYTE DESCRIPTION
986 NUMB   IN     OUT    IN
987 -----
988 PORT   RD-STA  WR-DAT  RD-DAT  Z80 I/O PORT
989 =====
990 00              0x41      = "A" (ASM COMND) *** DIRECT I/O CMD TO THE TS-PICO ***
991 01              0x50      = SUBSYSTEM ID (50 = FILE SYSTEM_CMD)
992 02              0x02      = FUNCTION ID (02 = OPEN)
993 03              0x03      = ACCESS_MODE (00=DEFAULT / 01=READ ONLY MODE /
994                               02=WRITE ONLY MODE / 03=READ/WRITE MODE )
995 04              0x01      = BLOCK_SIZE (00=USER DEFINED BLOCK / 01=STD 256 BYTES /
996                               02=512 BYTES / 04=1024 / 08=2048 /16=4096 /32=8192 /64=16384)
997 05              0x00      = USER ID (SET BY THE USER (00=AUTOMATIC ASIGNMENT / USER ID VALUE>00)
998 06              0x00      = PMR_1 USER DEFINED BLOCK SIZE LSB (USED ONLY IN BLOCK_SIZE==00)
999 07              0x00      = PMR_2 USER DEFINED BLOCK SIZE MSB (USED ONLY IN BLOCK_SIZE==00)
1000 08              0x00      = MODE (00=BINARY / 01=ASCII / 02=ASCII EXTENDED /
1001                               03 = LINE BY LINE CR / 04 = LINE BY LINE CR+LF)
1002 -----
1003 09              0x11      CRC FROM TIMEX (from bytes 00 to 08)
1004 -----
1005 10              0xA1      Read Status Code (A1 = No Error)
1006 -----
1007 -----WAIT EXECUTION 2.8ms-----
1008 -----
1009 11              0x40      Read CONTINUE FLAG (BIT 6=1)
1010 -----
1011 12              0x44      "D" (DATA BLOCK CMD)
1012 13              0x0A      COMND LEN L
1013 14              0x00      COMND LEN H
1014 -----
1015 15              0x43      "C"
1016 16              0x50      "P"
1017 17              0x4D      "M"
1018 18              0x2D      "-"
1019 19              0x32      "2"
1020 20              0x32      "2"
1021 21              0x2E      "."
1022 22              0x49      "I"
1023 23              0x4D      "M"
1024 24              0x47      "G"
1025 -----
1026 25              0x50      CRC FROM TIMEX
1027 -----
1028 26              0xA1      Read Status Code (A1 = No Error)
1029 -----
1030 -----WAIT EXECUTION 2.8ms-----
1031 -----
1032 27              0x40      Read CONTINUE FLAG (BIT 6=1)
1033 -----
1034 28              0x44      "D" (RESPONSE TO PARAM. BLOCK)
1035 29              0x08      PARAM BLOCK LEN L
1036 30              0x00      PARAM BLOCK LEN H
1037 -----
1038 31              0xA1      COMMAND EXECUTION STATUS (0xA1 = OK - NO ERROR)
1039 -----
1040 32              0x2B      DATA PARAMETER OPEN ID HEADER ( 2A = ID HEADER )
1041 33              0x01      SESION ID (ASSIGNED AUTOMATICALLY BY THE SYSTEM)
1042 34              0x01      USER_ID (USER DEFINED VALUE FOR THIS FILE)
1043 -----
1044 35              0x2C      DATA PARAMETER FILESIZE IN KB ( 2B = FILESIZE )
1045 36              0x01      FILESIZE LOW BYTE
1046 37              0x01      FILESIZE MID BYTE
1047 38              0x01      FILESIZE HIGH BYTE
1048 -----
1049 39              0xEB      CRC FROM TS-PICO (from bytes 28 to 34)
1050 -----
1051 40              0x01      Send END Command / ERROR CODE FROM TIMEX
1052 -----
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1053 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 15

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1055

1056 READ BLOCK COMMAND = 03 - READ A BLOCK FROM AN OPEN FILE

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1059

1060 BYTE STAT DATA DATA BYTE DESCRIPTION

1061 NUMB IN OUT IN

1062

1063 PORT RD-STA WR-DAT RD-DAT Z80 I/O PORT

1064

1065 00 0x41 = "A" (ASM COMND) *** DIRECT I/O CMD TO THE TS-PICO ***

1066 01 0x50 = SUBSYSTEM ID (50 = FILE_SYSTEM_CMD)

1067 02 0x03 = FUNCTION ID (03 = READ_BLOCK)

1068 03 0x01 = READ_MODE (00 = NORMAL / 01 = READ FROM FILE /

1069 02 = READ FROM POINTER / 03 = READ FROM CACHE DATA)

1070 04 0x01 = SESION ID (SYSTEM ASSIGN THIS VALUE, RETURNED ON OPEN FUNC)

1071 05 0x01 = USER ID (VALUE SET BY THE CALLER)

1072 06 0x00 = PMR_1 READ BLOCK NUMBER LSB

1073 07 0x00 = PMR_2 READ BLOCK NUMBER MSB

1074 08 0x00 = PMR_3 READ EXTENT NUMBER

1075

1076 09 0x13 CRC FROM TIMEX (from bytes 00 to 08)

1077

1078 10 0xA1 Read Status Code (A1 = No Error)

1079

1080 -----WAIT EXECUTION 2.8ms-----

1081

1082 11 0x40 Read CONTINUE FLAG (BIT 6=1)

1083

1084 12 0x44 "D" (DATA BLOCK CMD)

1085 13 0x01 BLOCK LEN L (BLOCK SIZE = 256 BYTES)

1086 14 0x01 BLOCK LEN H (IF READ ERROR BLOCK SIZE = 000 BYTES)

1087

1088 15 0xA1 COMMAND EXECUTION STATUS (0xA1 = OK - NO ERROR)

1089

1090 16 0x00 DATA [000]

1091 17 0x00 DATA [001]

1092 .

1093 .

1094 . *** ALL DATA RETURN 0x00 ***

1095

1096 270 0x00 DATA [254]

1097 271 0x00 DATA [255]

1098

1099 272 0xE5 CRC FROM TS-PICO (from bytes 12 to 271)

1100

1101 273 0x01 Send END Command / ERROR CODE FROM TIMEX

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1190 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 17

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SEEK COMMAND = 05

- MOVE ACCESS POINTER IN A OPEN FILE

BYTE	STAT	DATA	DATA	BYTE DESCRIPTION
NUMB	IN	OUT	IN	

PORT	RD-STA	WR-DAT	RD-DAT	Z80 I/O PORT
------	--------	--------	--------	--------------

00		0x41		= "A" (ASM COMND) *** DIRECT I/O CMD TO THE TS-PICO ***
01		0x50		= SUBSYSTEM ID (50 = FILE_SYSTEM_CMD)
02		0x05		= FUNCTION ID (05 = SEEK)
03		0x01		= SEEK MODE (00=GET POINTER / 01=SET POINTER / 02=SET TO BEGIN / 03=SET TO END)
04		0x01		= SESION ID (SYSTEM ASSIGN THIS FIELD AS A RANDOM VALUE RETURNED ON OPEN FUNC)
05		0x01		= USER ID (SET BY THE CALLER)
06		0x40		= PMR_1 SEEK BLOCK NUMBER LSB (ONLY USED BY 01=SET POINTER MODE)
07		0x00		= PMR_2 SEEK BLOCK NUMBER MSB (ONLY USED BY 01=SET POINTER MODE)
08		0x00		= PMR_3 SEEK EXTENT NUMBER (ONLY USED BY 01=SET POINTER MODE STARTING FROM 00h)

09		0x55		CRC FROM TIMEX (from bytes 00 to 08)
----	--	------	--	--------------------------------------

10			0xA1	Read Status Code (A1 = No Error)
----	--	--	------	----------------------------------

-----WAIT EXECUTION 2.8ms-----

11		0x40		Read CONTINUE FLAG (BIT 6=1)
----	--	------	--	------------------------------

12			0x44	"D" (DATA BLOCK CMD)
----	--	--	------	----------------------

13			0x07	BLOCK LEN L
----	--	--	------	-------------

14			0x00	BLOCK LEN H (BLOCK SIZE = 256 BYTES)
----	--	--	------	--------------------------------------

15			0xA1	COMMAND EXECUTION STATUS (0xA1 = OK - NO ERROR)
----	--	--	------	---

16			0x00	PREVIOUS SEEK BLOCK NUMBER LSB
----	--	--	------	--------------------------------

17			0x00	PREVIOUS SEEK BLOCK NUMBER MSB
----	--	--	------	--------------------------------

18			0x00	PREVIOUS SEEK EXTENT NUMBER
----	--	--	------	-----------------------------

19			0x40	NEW SEEK BLOCK LOCATION NUMBER LSB
----	--	--	------	------------------------------------

20			0x00	NEW SEEK BLOCK LOCATION NUMBER MSB
----	--	--	------	------------------------------------

21			0x00	NEW SEEK EXTENT NUMBER
----	--	--	------	------------------------

22			0xA2	CRC FROM TS-PICO (from bytes 12 to 21)
----	--	--	------	--

23			0x01	Send END Command / ERROR CODE FROM TIMEX
----	--	--	------	--

1260 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 18

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1261 -----
1262 CLOSE COMMAND = 06          - CLOSE AN OPEN FILE
1263 -----
1264
1265 =====
1266 BYTE      STAT   DATA   DATA   BYTE DESCRIPTION
1267 NUMB      IN     OUT     IN
1268 ----      -
1269 PORT      RD-STA  WR-DAT  RD-DAT  Z80 I/O PORT
1270 -----
1271
1272 00          0x41          = "A" (ASM COMND) *** DIRECT I/O CMD TO THE TS-PICO ***
1273 01          0x50          = SUBSYSTEM ID (50 = FILE_SYSTEM_CMD)
1274 02          0x06          = FUNCTION ID (06 = CLOSE)
1275 03          0x00          = CLOSE MODE (00=DEFAULT / 01=FORCE CLOSE)
1276 04          0x01          = SESION ID (SYSTEM ASSIGN THIS FIELD ON OPEN FUNC)
1277 05          0x01          = USER ID USER ID VALUE (SET BY THE BIOS)
1278 06          0x00          = PMR_1 NOT USED
1279 07          0x00          = PMR_2 NOT USED
1280 08          0x00          = PMR_3 NOT USED
1281 -----
1282 09          0x17          CRC FROM TIMEX (from bytes 00 to 08)
1283 -----
1284 10          0xA1          Read Status Code (A1 = No Error)
1285 -----
1286 -----WAIT EXECUTION 2.8ms-----
1287 -----
1288 11          0x40          Read CONTINUE FLAG (BIT 6=1)
1289 -----
1290 12          0x44          "D" (RESPONSE TO PARAM. BLOCK)
1291 13          0x01          PARAM BLOCK LEN L
1292 14          0x00          PARAM BLOCK LEN H
1293 -----
1294 15          0xA1          COMMAND EXECUTION STATUS (0xA1 = OK - NO ERROR)
1295 -----
1296 16          0xE4          CRC FROM TS-PICO (from bytes 12 to 15)
1297 -----
1298 17          0x01          Send END Command / ERROR CODE FROM TIMEX
1299 -----
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1330 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 19

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1333 CREATE FILE COMMAND = 07 - CREATE A NEW FILE (DO NOT OPEN THE FILE FOR R/W)

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1400 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 20
1401
1402 -----
1403 ERASE FILE COMMAND = 08          - ERASE A CLOSE FILE FROM STORAGE
1404 -----
1405
1406 =====
1407 BYTE   STAT   DATA   DATA   BYTE DESCRIPTION
1408 NUMB   IN     OUT     IN
1409 ----   -
1410 PORT   RD-STA WR-DAT  RD-DAT  Z80 I/O PORT
1411 =====
1412 00                0x41      = "A" (ASM COMND) *** DIRECT I/O CMD TO THE TS-PICO ***
1413 01                0x50      = SUBSYSTEM ID (50 = FILE_SYSTEM_CMD)
1414 02                0x08      = FUNCTION ID (08 = ERASE)
1415 03                0x01      = ERASE_MODE (01 = ERASE FILE / 02 = ERASE DIRECTORY)
1416 04                0x00      = RESERVED NOT USED
1417 05                0x00      = PMR_0 NOT USED
1418 06                0x00      = PMR_1 NOT USED
1419 07                0x00      = PMR_2 NOT USED
1420 08                0x00      = PMR_3 NOT USED
1421 -----
1422 09                0x18      CRC FROM TIMEX (from bytes 00 to 08)
1423 -----
1424 10                0xA1      Read Status Code (A1 = No Error)
1425 -----
1426 -----WAIT EXECUTION 2.8ms-----
1427 -----
1428 11                0x40      Read CONTINUE FLAG (BIT 6=1)
1429 -----
1430 12                0x44      "D" (DATA BLOCK CMD)
1431 13                0x0A      COMND LEN L
1432 14                0x00      COMND LEN H
1433 -----
1434 15                0x43      "C"
1435 16                0x50      "p"
1436 17                0x4D      "M"
1437 18                0x2D      "L"
1438 19                0x32      "2"
1439 20                0x32      "2"
1440 21                0x2E      "."
1441 22                0x49      "I"
1442 23                0x4D      "M"
1443 24                0x47      "G"
1444 -----
1445 25                0x50      CRC FROM TIMEX
1446 -----
1447 -----WAIT EXECUTION 2.8ms-----
1448 -----
1449 26                0x40      Read CONTINUE FLAG (BIT 6=1)
1450 -----
1451 27                0x44      "D" (RESPONSE TO PARAM. BLOCK)
1452 28                0x01      PARAM BLOCK LEN L
1453 29                0x00      PARAM BLOCK LEN H
1454 -----
1455 30                0xA1      COMMAND EXECUTION STATUS (0xA1 = OK - NO ERROR)
1456 -----
1457 31                0xE4      CRC FROM TS-PICO (from bytes 27 to 30)
1458 -----
1459 32                0x01      Send END Command / ERROR CODE FROM TIMEX
1460 -----
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1470 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 21
1471
1472 -----
1473 CREATE DIRECTORY COMMAND = 09          - CREATE A DIRECTORY FOLDER
1474                                         (RETURN THE PATH OF THE CURRENT DIRECTORY)
1475 -----
1476
1477 =====
1478 BYTE   STAT   DATA   DATA   BYTE DESCRIPTION
1479 NUMB   IN     OUT     IN
1480 ----   -
1481 PORT   RD-STA WR-DAT  RD-DAT  Z80 I/O PORT
1482 -----
1483 00      0x41      = "A" (ASM COMND) *** DIRECT I/O CMD TO THE TS-PICO ***
1484 01      0x50      = SUBSYSTEM_ID (50 = FILE_SYSTEM_CMD)
1485 02      0x09      = FUNCTION_ID (09 = CREATE DIRECTORY)
1486 03      0x00      = CREATE_MODE (00 = FROM CURRENT LOCATION / 01=FROM ROOT)
1487 04      0x00      = RESERVED      NOT USED
1488 05      0x00      = PMR_0         NOT USED
1489 06      0x00      = PMR_1         NOT USED
1490 07      0x00      = PMR_2         NOT USED
1491 08      0x00      = PMR_3         NOT USED
1492 -----
1493 09      0x18      CRC FROM TIMEX (from bytes 00 to 08)
1494 -----
1495 10      0xA1      Read Status Code (A1 = No Error)
1496 -----
1497 -----WAIT EXECUTION 2.8ms-----
1498 -----
1499 11      0x40      Read CONTINUE FLAG (BIT 6=1)
1500 -----
1501 12      0x44      "D" (DATA BLOCK CMD)
1502 13      0x0A      COMND LEN L
1503 14      0x00      COMND LEN H
1504 -----
1505 15      0x43      "C"
1506 16      0x50      "P"
1507 17      0x4D      "M"
1508 18      0x2D      "-"
1509 19      0x44      "D"
1510 20      0x41      "A"
1511 21      0x54      "T"
1512 22      0x41      "A"
1513 23      0x30      "0"
1514 24      0x31      "1"
1515 -----
1516 25      0x2C      CRC FROM TIMEX
1517 -----
1518 -----WAIT EXECUTION 2.8ms-----
1519 -----
1520 26      0x40      Read CONTINUE FLAG (BIT 6=1)
1521 -----
1522 27      0x44      "D" (RESPONSE TO PARAM. BLOCK)
1523 28      0x06      PARAM BLOCK LEN L
1524 29      0x00      PARAM BLOCK LEN H
1525 -----
1526 30      0xA1      COMMAND EXECUTION STATUS (0xA1 = OK - NO ERROR)
1527 -----
1528 31      0x04      CURRENT_PATH_LEN ( 00 = NO MOUNTED FILE)
1529 32      0x5C      "\" CURRENT DIRECTORY
1530 33      0x44      "D" or 00 (NULL) IF NOT INFO AVAILABLE
1531 34      0x53      "S"
1532 35      0x4B      "K"
1533 -----
1534 36      0xE7      CRC FROM TS-PICO (from bytes 27 to 35)
1535 -----
1536 37      0x01      Send END Command / ERROR CODE FROM TIMEX
1537 -----
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1540 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 22

1541

1542

1543 CHANGE DIRECTORY COMMAND = 10 (RETURN THE PATH OF THE CURRENT DIRECTORY)

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1545

1546

1547 BYTE STAT DATA DATA BYTE DESCRIPTION

1548 NUMB IN OUT IN

1549

1550 PORT RD-STA WR-DAT RD-DAT Z80 I/O PORT

1551

1552 00 0x41 = "A" (ASM COMND) *** DIRECT I/O CMD TO THE TS-PICO ***

1553 01 0x50 = SUBSYSTEM ID (50 = FILE_SYSTEM_CMD)

1554 02 0x0A = FUNCTION ID (10 = CHANGE DIRECTORY)

1555 03 0x01 = FUNC. MODE (00=GET CURRENT LOCATION / 01=START FROM CURRENT DIRECTORY / 02=START FROM ROOT)

1556 04 0x00 = RESERVED NOT USED

1557 05 0x00 = PMR_0 NOT USED

1558 06 0x00 = PMR_1 NOT USED

1559 07 0x00 = PMR_2 NOT USED

1560 08 0x00 = PMR_3 NOT USED

1561

1562 09 0x1A CRC FROM TIMEX (from bytes 00 to 08)

1563

1564 10 0xA1 Read Status Code (A1 = No Error)

1565

1566 -----WAIT EXECUTION 2.8ms-----

1567

1568 11 0x40 Read CONTINUE FLAG (BIT 6=1)

1569

1570 12 0x44 "D" (DATA BLOCK CMD)

1571 13 0x0A COMND LEN L

1572 14 0x00 COMND LEN H

1573

1574 15 0x43 "C"

1575 16 0x50 "P"

1576 17 0x4D "M"

1577 18 0x2D "_"

1578 19 0x44 "D"

1579 20 0x41 "A"

1580 21 0x54 "T"

1581 22 0x41 "A"

1582 23 0x30 "0"

1583 24 0x31 "1"

1584

1585 25 0x2C CRC FROM TIMEX

1586

1587 -----WAIT EXECUTION 2.8ms-----

1588

1589 26 0x40 Read CONTINUE FLAG (BIT 6=1)

1590

1591 27 0x44 "D" (RESPONSE TO PARAM. BLOCK)

1592 28 0x07 PARAM BLOCK LEN L

1593 29 0x00 PARAM BLOCK LEN H

1594

1595 30 0xA1 COMMAND EXECUTION STATUS (0xA1 = OK - NO ERROR)

1596

1597 31 0x0F CURRENT_PATHNAME_LEN (00 = IF NO FILE SYSTEM READY)

1598 32 0x5C "\" or 00 (NULL) IF NOT INFO AVAILABLE

1599 33 0x44 "D"

1600 34 0x53 "S"

1601 35 0x4B "K"

1602 36 0x5C "\"

1603

1604 37 0xB1 CRC FROM TS-PICO (from bytes 27 to 36)

1605

1606 38 0x01 Send END Command / ERROR CODE FROM TIMEX

1607

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1610 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 23

1611
1612 -----
1613 CURRENT DIRECTORY LISTING COMMAND = 11 (RETURN THE CURRENT DIRECTORY - 63 COL/LINE)
1614 -----

1615
1616 *****
1617 *** DIRECTORY EXAMPLE HEX DUMP ***
1618 *****

1619 \DSK
1620 -----
1621 CPM-22.IMG 6,836,224 07/01/2023 09:30 PM
1622 Downloads <DIR> 04/02/2024 10:01 PM
1623 Favorites <DIR> 07/04/2021 02:21 PM
1624 -----
1625 0000 43 50 4D 2D 32 32 2E 49 4D 47 20 20 20 20 20 20 CPM-22.IMG
1626 0010 20 20 20 20 20 20 20 20 20 20 36 2C 38 33 36 2C 32 6,836,2
1627 0020 32 34 20 20 20 20 20 20 20 30 37 2F 30 31 2F 32 24 07/01/2
1628 0030 30 32 33 20 20 30 39 3A 33 30 20 50 4D 20 0D 0A 023 09:30 PM ..
1629 0040 44 6F 77 6E 6C 6F 61 64 73 20 20 20 20 20 20 20 Downloads
1630 0050 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
1631 0060 20 20 20 3C 44 49 52 3E 20 30 34 2F 30 32 2F 32 <DIR> 04/02/2
1632 0070 30 32 34 20 20 31 30 3A 30 31 20 50 4D 20 0D 0A 024 10:01 PM ..
1633 0080 46 61 76 6F 72 69 74 65 73 20 20 20 20 20 20 20 Favorites
1634 0090 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
1635 00A0 20 20 20 3C 44 49 52 3E 20 30 37 2F 30 34 2F 32 <DIR> 07/04/2
1636 00B0 30 32 31 20 20 30 32 3A 32 31 20 50 4D 20 0D 0A 021 02:21 PM ..
1637 -----

1638
1639 =====

BYTE	STAT	DATA	DATA	BYTE DESCRIPTION
NUMB	IN	OUT	IN	
----	----	----	----	
PORT	RD-STA	WR-DAT	RD-DAT	Z80 I/O PORT
00		0x41		= "A" (ASM COMND) *** DIRECT I/O CMD TO THE TS-PICO ***
01		0x50		= SUBSYSTEM ID (50 = FILE SYSTEM CMD)
02		0x0B		= FUNCTION ID (11 = CURRENT DIRECTORY LISTING)
03		0x01		= FUNC. MODE (00=GET CURRENT DIR STATUS / 01=SHOW FILES IN CURRENT DIR ONLY / 02=SHOW DIR ONLY / 03= ALL DIR ENTRIES)
04		0x01		= LIST FMT (00=ASCII+CR / 01=ASCII+CR+LF / 04=ASCII+CR+LF+NULL / 08 = NULL)
05		0x00		= PMR_0 NOT USED
06		0x00		= PMR_1 NOT USED
07		0x00		= PMR_2 NOT USED
08		0x00		= PMR_3 NOT USED

1654
1655 09 0x1A CRC FROM TIMEX (from bytes 00 to 08)
1656 -----

1657 10 0xA1 Read Status Code (01 = No Error)
1658 -----

1659 -----WAIT EXECUTION 2.8ms-----
1660 -----

1661 11 0x40 Read CONTINUE FLAG (BIT 6=1)
1662 -----

1663 12 0x44 "D" (DATA BLOCK CMD)
1664 13 0x0C COMND LEN L
1665 14 0x00 COMND LEN H
1666 -----

1667 15 0x01 DIR FMT (00=DEFAULT / 01=64_COL / 02=32_COL / 03=ONLY NAMES / 04=ONLY DIRs)
1668 16 0x0A NAME_LEN (00 = IF NO FILE NAME - ALL -> *.*.)
1669 17 0x43 "C"
1670 18 0x50 "P" Directory SEEK Text
1671 19 0x4D "M" formatted in
1672 20 0x2D "L" 31 Cols Format
1673 21 0x32 "2" Single Row Per File
1674 22 0x32 "2"
1675 23 0x2E "."
1676 24 0x49 "I"
1677 25 0x4D "M"
1678 26 0x47 "G"
1679 -----

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1680 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 24

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1681
1682 =====
1683 BYTE      STAT  DATA  DATA  BYTE DESCRIPTION
1684 NUMB      IN    OUT   IN
1685 ----
1686 PORT      RD-STA WR-DAT  RD-DAT  Z80 I/O PORT
1687 =====
1688 27          0x5D          CRC FROM TIMEX
1689 -----
1690 -----WAIT EXECUTION 2.8ms-----
1691 -----
1692 28          0x40          Read CONTINUE FLAG (BIT 6=1)
1693 -----
1694 29          0x44          "D" (RESPONSE TO PARAM. BLOCK)
1695 30          0x01          PARAM BLOCK LEN L
1696 31          0x00          PARAM BLOCK LEN H
1697 -----
1698 32          0xA1          COMMAND EXECUTION STATUS (0xA1 = OK - NO ERROR)
1699 -----
1700 33          0x01          CURRENT_DIR_ENTRIES LSB ( 0000 = NO DIRECTORY ENTRIES)
1701 34          0x00          CURRENT_DIR_ENTRIES MSB
1702 35          0x5C          "\" DIRECTORY TEXT INFORMATION
1703 36          0x44          "D" or 00 (NULL) IF NOT INFO AVAILABLE
1704 37          0x53          "S"
1705 38          0x4B          "K" FIRST LINE OF DIRECTORY IS THE PATHNAME
1706 -----
1707 39          0x01          CURRENT_LIST_DIR_ENTRIES LSB ( 0000 = NO DIRECTORY ENTRIES)
1708 40          0x00          CURRENT_LIST_DIR_ENTRIES MSB
1709 -----
1710 41          0x0A          CURRENT_NAME_LEN (00 = IF NO FILE OPENED)
1711 42          1            0x43          "C"
1712 43          2            0x50          "P"
1713 44          3            0x4D          "M"
1714 45          4            0x2D          "-"
1715 46          5            0x32          "2"
1716 47          6            0x32          "2"
1717 48          7            0x2E          "."
1718 49          8            0x49          "I"
1719 50          9            0x4D          "M"
1720 51          10           0x47          "G"
1721 52          11           0x20          " "
1722 53          12           0x20          " "
1723 54          13           0x20          " "
1724 55          14           0x20          " "
1725 56          15           0x20          " "
1726 57          16           0x20          " "
1727 58          17           0x20          " "
1728 59          18           0x20          " "
1729 60          19           0x20          " "
1730 61          20           0x20          " "
1731 62          21           0x20          " "
1732 63          22           0x20          " "
1733 64          23           0x20          " "
1734 65          24           0x20          " "
1735 66          25           0x20          " "
1736 67          26           0x36          "6"
1737 68          27           0x2C          ","
1738 69          28           0x38          "8"
1739 70          29           0x33          "3"
1740 71          30           0x36          "6"
1741 72          31           0x2C          ","
1742 73          32           0x32          "2"
1743 74          33           0x32          "2"
1744 75          34           0x34          "4"
1745 76          35           0x20          " "
```

1750 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 25

```
1751
1752 =====
1753 BYTE      STAT      DATA      DATA      BYTE DESCRIPTION
1754 NUMB      IN        OUT        IN
1755 ----      -
1756 PORT      RD-STA    WR-DAT      RD-DAT      Z80 I/O PORT
1757 =====
1758 77          36        0x20        " "
1759 78          37        0x20        " "
1760 79          38        0x20        " "
1761 80          39        0x20        " "
1762 81          40        0x20        " "
1763 82          41        0x20        " "
1764 83          42        0x30        "0"
1765 84          43        0x37        "7"
1766 85          44        0x2F        "/"
1767 86          45        0x30        "0"
1768 87          46        0x31        "1"
1769 88          47        0x2F        "/"
1770 89          48        0x32        "2"
1771 90          49        0x30        "0"
1772 91          50        0x32        "2"
1773 92          51        0x33        "3"
1774 93          52        0x20        " "
1775 94          53        0x20        " "
1776 95          54        0x30        "0"
1777 96          55        0x39        "9"
1778 97          56        0x3A        ":"
1779 98          57        0x33        "3"
1780 99          58        0x30        "0"
1781 100         59        0x20        " "
1782 101         60        0x50        "P"
1783 102         61        0x4D        "M"
1784 103         62        0x20        " "
1785 104         63        0x0D        <ENTER> (OPTIONAL 0Dh+0Ah / 0x00 / 0Dh+0Ah+00h )
1786 105         64        0x0A        <LINE FEED>
1787 -----
1788 106          0xF0      CRC FROM TS-PICO      (from bytes 27 to 30)
1789 -----
1790 107          0x01          Send END Command / ERROR CODE FROM TIMEX
1791 -----
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1820 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 26

```
1821 -----
1822 WAIT FOR MOUNT MEDIA COMMAND = 12          - MOUNT MEDIA - PREPARE TO INSERT MEDIA
1823 -----
1824 -----
1825 -----
```

```
1826 =====
1827 BYTE   STAT   DATA   DATA   BYTE DESCRIPTION
1828 NUMB   IN     OUT     IN
1829 ----   -
1830 PORT   RD-STA  WR-DAT  RD-DAT  Z80 I/O PORT
1831 =====
1832 00              0x41      = "A" (ASM COMND) *** DIRECT I/O CMD TO THE TS-PICO ***
1833 01              0x50      = SUBSYSTEM ID (50 = FILE_SYSTEM_CMD)
1834 02              0x0C      = FUNCTION ID (0C = INSERT MEDIA)
1835 03              0x00      = INSERT MODE (00=DEFAULT GET INFO / 01=GET MOUNT STATUS / 02=WAIT TILL INSERT)
1836 04              0x00      = RESERVED NOT USED
1837 05              0x00      = PMR_0 NOT USED
1838 06              0x00      = PMR_1 NOT USED
1839 07              0x00      = PMR_2 NOT USED
1840 08              0x00      = PMR_3 NOT USED
1841 -----
1842 09              0x1D      CRC FROM TIMEX (from bytes 00 to 08)
1843 -----
1844 10              0xA1      Read Status Code (A1 = No Error)
1845 -----
1846 -----WAIT EXECUTION 2.8ms-----
1847 -----
1848 11              0x40      Read CONTINUE FLAG (BIT 6=1)
1849 -----
1850 12              0x44      "D" (RESPONSE TO PARAM. BLOCK)
1851 13              0x01      PARAM BLOCK LEN L
1852 14              0x00      PARAM BLOCK LEN H
1853 -----
1854 15              0xA1      COMMAND EXECUTION STATUS (0xA1 = OK - NO ERROR)
1855 -----
1856 16              0x2A      DATA PARAMETER MOUNT INFO SIZE IN KB ( 2B = FILESIZE )
1857 17              0x00      MOUNT_FLAG
1858 18              0x00      MOUNT_SIZE_LOW (LSB = SIZE IN MEGABYTES)
1859 19              0x00      MOUNT_SIZE_HIGH (LSB = SIZE IN MEGABYTES)
1860 20              0x00      MOUNT_MEDIA_TYPE ( 00 = NA / 01=SD / 02=RAM / 03=SSD / 04=IP-SERVER)
1861 -----
1862 21              0x00      MOUNT_INFO_ASCII_FLAG ( 00 = NONE)
1863 -----
1864 22              0xE1      CURRENT_DISK_INFO_LEN_LSB (00 = IF NO INFO AVAILABLE)
1865 23              0x00      CURRENT_DISK_INFO_LEN_MSB
1866 -----
1867 24              0x43      "C"
1868 25              0x61      "a"
1869 26              0x72      "r"
1870 27              0x64      "d"
1871 28              0x20      " "
1872 29              0x69      "i"
1873 30              0x73      "s"
1874 31              0x20      " "
1875 32              0x70      "p"
1876 33              0x72      "r"
1877 34              0x65      "e"
1878 35              0x73      "s"
1879 36              0x65      "e"
1880 37              0x6E      "n"
1881 38              0x74      "t"
1882 39              0x2E      "."
1883 40              0x0D      " "
1884 41              0x0A      " "
1885 42              0x43      "C"
1886 43              0x61      "a"
1887 44              0x72      "r"
1888 45              0x64      "d"
1889
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```

1890 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 27

1891	=====				
1892	BYTE	STAT	DATA	DATA	BYTE DESCRIPTION
1893	NUMB	IN	OUT	IN	
1894	----	----	----	----	
1895	PORT	RD-STA	WR-DAT	RD-DAT	Z80 I/O PORT
1896	=====				
1897	46			0x20	" "
1898	47			0x74	"t"
1899	48			0x79	"y"
1900	49			0x70	"p"
1901	50			0x65	"e"
1902	51			0x3A	": "
1903	52			0x20	" "
1904	53			0x20	" "
1905	54			0x20	" "
1906	55			0x20	" "
1907	56			0x20	" "
1908	57			0x20	" "
1909	58			0x20	" "
1910	59			0x20	" "
1911	60			0x20	" "
1912	61			0x53	"S"
1913	62			0x44	"D"
1914	63			0x48	"H"
1915	64			0x43	"C"
1916	65			0x0D	" "
1917	66			0x0A	" "
1918	67			0x43	"C"
1919	68			0x6C	"l"
1920	69			0x75	"u"
1921	70			0x73	"s"
1922	71			0x74	"t"
1923	72			0x65	"e"
1924	73			0x72	"r"
1925	74			0x73	"s"
1926	75			0x3A	": "
1927	76			0x20	" "
1928	77			0x20	" "
1929	78			0x20	" "
1930	79			0x20	" "
1931	80			0x20	" "
1932	81			0x20	" "
1933	82			0x20	" "
1934	83			0x20	" "
1935	84			0x20	" "
1936	85			0x20	" "
1937	86			0x34	"4"
1938	87			0x38	"8"
1939	88			0x35	"5"
1940	89			0x39	"9"
1941	90			0x33	"3"
1942	91			0x36	"6"
1943	92			0x0D	" "
1944	93			0x0A	" "
1945	94			0x42	"B"
1946	95			0x6C	"l"
1947	96			0x6F	"o"
1948	97			0x63	"c"
1949	98			0x6B	"k"
1950	99			0x73	"s"
1951	100			0x20	" "
1952	101			0x78	"x"
1953	102			0x20	" "
1954	103			0x43	"C"
1955	104			0x6C	"l"
1956	105			0x75	"u"
1957	106			0x73	"s"
1958					
1959					
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1960					

2010 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 29

2011	=====				
2012	BYTE	STAT	DATA	DATA	BYTE DESCRIPTION
2013	NUMB	IN	OUT	IN	
2014	----	----	----	----	
2015	PORT	RD-STA	WR-DAT	RD-DAT	Z80 I/O PORT
2016	=====				
2017	146			0x42	"B"
2018	147			0x6C	"l"
2019	148			0x6F	"o"
2020	149			0x63	"c"
2021	150			0x6B	"k"
2022	151			0x20	" "
2023	152			0x53	"S"
2024	153			0x69	"i"
2025	154			0x7A	"z"
2026	155			0x65	"e"
2027	156			0x3A	":"
2028	157			0x20	" "
2029	158			0x20	" "
2030	159			0x20	" "
2031	160			0x20	" "
2032	161			0x20	" "
2033	162			0x20	" "
2034	163			0x20	" "
2035	164			0x20	" "
2036	165			0x35	"5"
2037	166			0x31	"1"
2038	167			0x32	"2"
2039	168			0x0D	" "
2040	169			0x0A	" "
2041	170			0x56	"V"
2042	171			0x6F	"o"
2043	172			0x6C	"l"
2044	173			0x75	"u"
2045	174			0x6D	"m"
2046	175			0x65	"e"
2047	176			0x20	" "
2048	177			0x74	"t"
2049	178			0x79	"y"
2050	179			0x70	"p"
2051	180			0x65	"e"
2052	181			0x20	" "
2053	182			0x69	"i"
2054	183			0x73	"s"
2055	184			0x3A	":"
2056	185			0x20	" "
2057	186			0x20	" "
2058	187			0x20	" "
2059	188			0x20	" "
2060	189			0x46	"F"
2061	190			0x41	"A"
2062	191			0x54	"T"
2063	192			0x33	"3"
2064	193			0x32	"2"
2065	194			0x0D	" "
2066	195			0x0A	" "
2067	196			0x56	"V"
2068	197			0x6F	"o"
2069	198			0x6C	"l"
2070	199			0x75	"u"
2071	200			0x6D	"m"
2072	201			0x65	"e"
2073	202			0x20	" "
2074					
2075					
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2080 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 30

2081	=====				
2082	BYTE	STAT	DATA	DATA	BYTE DESCRIPTION
2083	NUMB	IN	OUT	IN	
2084	----	----	----	----	
2085	PORT	RD-STA	WR-DAT	RD-DAT	Z80 I/O PORT
2086	=====				
2087	203			0x73	"s"
2088	204			0x69	"i"
2089	205			0x7A	"z"
2090	206			0x65	"e"
2091	207			0x20	" "
2092	208			0x28	"("
2093	209			0x4D	"M"
2094	210			0x62	"b"
2095	211			0x29	")"
2096	212			0x3A	":"
2097	213			0x20	" "
2098	214			0x20	" "
2099	215			0x31	"1"
2100	216			0x35	"5"
2101	217			0x31	"1"
2102	218			0x38	"8"
2103	219			0x35	"5"
2104	220			0x0D	" "
2105	221			0x0A	" "
2106	222			0x56	"v"
2107	223			0x6F	"o"
2108	224			0x6C	"l"
2109	225			0x75	"u"
2110	226			0x6D	"m"
2111	227			0x65	"e"
2112	228			0x20	" "
2113	229			0x46	"F"
2114	230			0x72	"r"
2115	231			0x65	"e"
2116	232			0x65	"e"
2117	233			0x20	" "
2118	234			0x53	"S"
2119	235			0x70	"p"
2120	236			0x61	"a"
2121	237			0x63	"c"
2122	238			0x65	"e"
2123	239			0x3A	":"
2124	240			0x20	" "
2125	241			0x31	"1"
2126	242			0x34	"4"
2127	243			0x32	"2"
2128	244			0x38	"8"
2129	245			0x30	"0"
2130	246			0x0D	<ENTER CR>
2131	247			0x0A	<LINE FEED>
2132	248			0x00	END MARKER
2133	-----				
2134	249			0x57	CRC FROM TS-PICO (from bytes 12 to 15)
2135	-----				
2136	250			0x01	Send END Command / ERROR CODE FROM TIMEX
2137	-----				
2138					
2139					
2140					
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2150 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 31

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REMOVE MEDIA COMMAND = 13 - UNMOUNT MEDIA - PREPARE TO POWER-DOWN

=====

BYTE STAT DATA DATA BYTE DESCRIPTION

NUMB IN OUT IN

----- -----

PORT RD-STA WR-DAT RD-DAT Z80 I/O PORT

=====

00 0x41 = "A" (ASM COMND) *** DIRECT I/O CMD TO THE TS-PICO ***

01 0x50 = SUBSYSTEM ID (50 = FILE_SYSTEM_CMD)

02 0x0D = FUNCTION ID (0D = REMOVE MEDIA)

03 0x00 = REMOVE MODE (00=DEFAULT / 01=FORCE REMOVE / 03=WAIT TILL REMOVE)

04 0x00 = RESERVED NOT USED

05 0x00 = PMR_0 NOT USED

06 0x00 = PMR_1 NOT USED

07 0x00 = PMR_2 NOT USED

08 0x00 = PMR_3 NOT USED

09 0x1C CRC FROM TIMEX (from bytes 00 to 08)

10 0xA1 Read Status Code (A1 = No Error)

-----WAIT EXECUTION 2.8ms-----

11 0x40 Read CONTINUE FLAG (BIT 6=1)

12 0x44 "D" (RESPONSE TO PARAM. BLOCK)

13 0x01 PARAM BLOCK LEN L

14 0x00 PARAM BLOCK LEN H

15 0xA1 COMMAND EXECUTION STATUS (0xA1 = OK - NO ERROR)

16 0xE4 CRC FROM TS-PICO (from bytes 12 to 15)

17 0x01 Send END Command / ERROR CODE FROM TIMEX

2220 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16 PAGE 32

2221
2222 *****
2223 * ADDITIONAL MEDIA SPECS *
2224 *****
2225

2226
2227 -----
2228 MEDIA CARD INFO SAMPLE SPECS
2229 -----
2230

2231 Card is present.
2232 Card type: SDHC
2233 Clusters: 485936
2234 Blocks x Cluster: 64
2235 Total Blocks: 31099904
2236 Block Size: 512
2237 Volume type is: FAT32
2238 Volume size (Mb): 15185
2239 Volume Free Space: 14280
2240

2241 -----
2242 HEX DUMP INFO
2243 -----
2244

2245 0000 43 61 72 64 20 69 73 20 70 72 65 73 65 6E 74 2E Card is present.
2246 0010 0D 0A 43 61 72 64 20 74 79 70 65 3A 20 20 20 20 ..Card type:
2247 0020 20 20 20 20 20 53 44 48 43 0D 0A 43 6C 75 73 74 SDHC..Clust
2248 0030 65 72 73 3A 20 20 20 20 20 20 20 20 20 20 20 20 ers: 48
2249 0040 35 39 33 36 0D 0A 42 6C 6F 63 6B 73 20 78 20 43 5936..Blocks x C
2250 0050 6C 75 73 74 65 72 3A 20 20 36 34 0D 0A 54 6F 74 luster: 64..Tot
2251 0060 61 6C 20 42 6C 6F 63 6B 73 3A 20 20 20 20 20 20 al Blocks:
2252 0070 33 31 30 39 39 39 30 34 0D 0A 42 6C 6F 63 6B 20 31099904..Block
2253 0080 53 69 7A 65 3A 20 20 20 20 20 20 20 20 20 20 20 20 Size: 512
2254 0090 0D 0A 56 6F 6C 75 6D 65 20 74 79 70 65 20 69 73 ..Volume type is
2255 00A0 3A 20 20 20 20 46 41 54 33 32 0D 0A 56 6F 6C 75 : FAT32..Volu
2256 00B0 6D 65 20 73 69 7A 65 20 28 4D 62 29 3A 20 20 31 me size (Mb): 1
2257 00C0 35 31 38 35 0D 0A 56 6F 6C 75 6D 65 20 46 72 65 5185..Volume Fre
2258 00D0 65 20 53 70 61 63 65 3A 20 31 34 32 38 30 0D 0A e Space: 14280..
2259
2260

2261
2262 <<<END OF DOC REV: 2.A>>>
2263

2264 LOW-LEVEL-ASSEMBLER EXTENSION FILE SYSTEM PROTOCOL-V2A 27/09/2023 15:16
2265

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2274
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